

RNN (SimpleRNN) Text Classifier — Toxic Category Detection

Task 1: Plain Bidirectional SimpleRNN trained on cellula_toxic_data.csv

1. Task Overview

The dataset contains 3,000 samples with a text 'query', an 'image descriptions' field, and a target label 'Toxic Category' with 9 classes: Child Sexual Exploitation, Elections, Non-Violent Crimes, Safe, Sex-Related Crimes, Suicide & Self-Harm, Unknown S-Type, Violent Crimes, unsafe. The query and image description were concatenated into a single text field and used to train a plain Recurrent Neural Network (Bidirectional SimpleRNN, without LSTM/GRU gating) to classify each sample into its correct toxic category.

Split	Samples
Train	2100
Validation	450
Test	450

2. Model Architecture

Embedding(vocab=3677, dim=128) → SpatialDropout1D(0.3) → Bidirectional SimpleRNN(64, return_sequences=True) → Bidirectional SimpleRNN(32) → Dense(64, ReLU) → Dropout(0.4) → Dense(9, Softmax). Trained with Adam (lr=1e-3), categorical cross-entropy loss, class weighting to counter class imbalance, and early stopping on validation loss. This model uses plain SimpleRNN cells (no gating mechanism), as required for Task 1.

3. Evaluation Results (Held-out Test Set)

Metric	Score
Accuracy	0.880
F1 Score (Weighted)	0.884
F1 Score (Macro)	0.902
Target F1	0.750

Result: F1 (weighted) = 0.884 ≥ target of 0.75 → PASSED

4. Per-Class Classification Report

precision recall f1-score support

```
Child Sexual Exploitation 1.000 1.000 1.000 15
Elections 0.944 1.000 0.971 17
Non-Violent Crimes 0.843 0.956 0.896 45
Safe 0.891 0.820 0.854 150
Sex-Related Crimes 1.000 0.941 0.970 17
Suicide & Self-Harm 1.000 1.000 1.000 17
Unknown S-Type 0.512 0.759 0.611 29
Violent Crimes 0.947 0.908 0.927 119
```

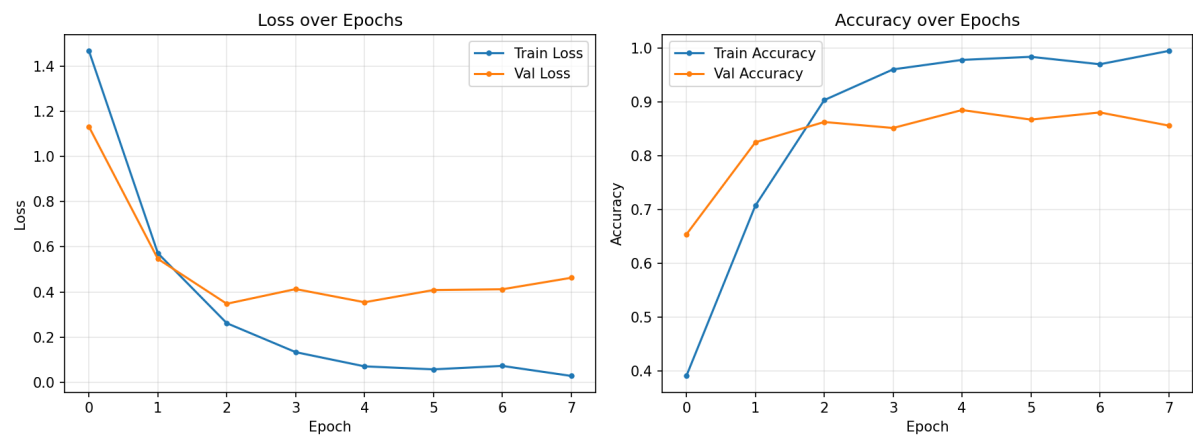
unsafe 0.921 0.854 0.886 41

accuracy 0.880 450

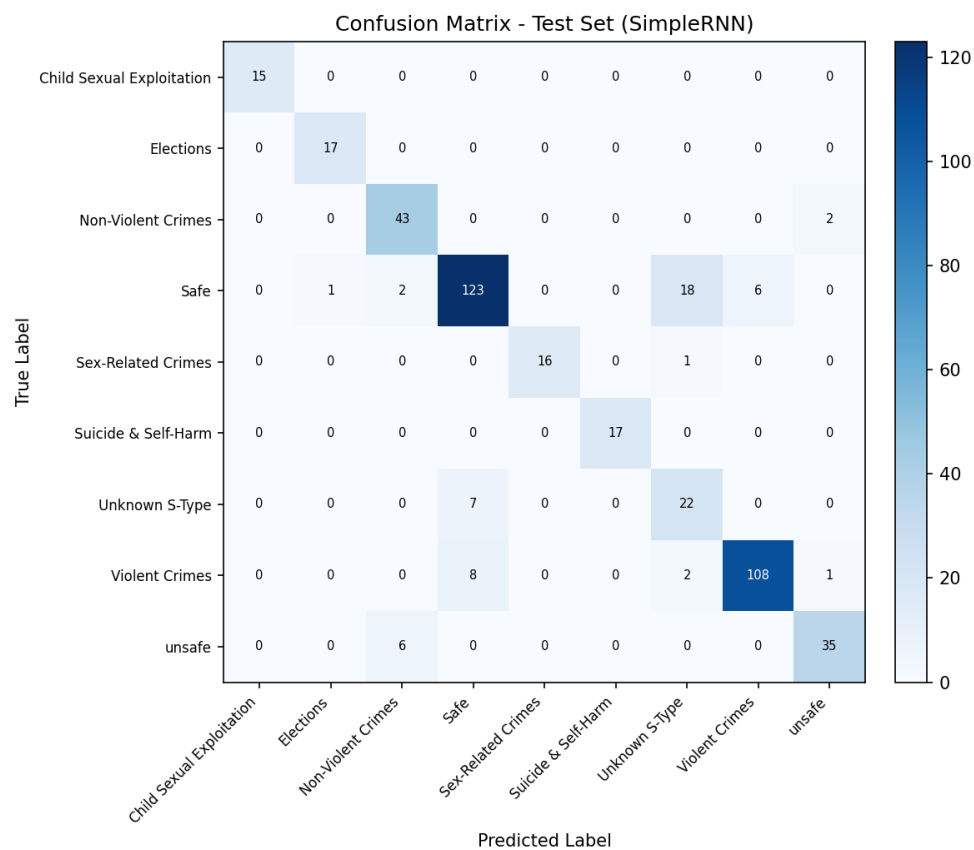
macro avg 0.895 0.915 0.902 450

weighted avg 0.893 0.880 0.884 450

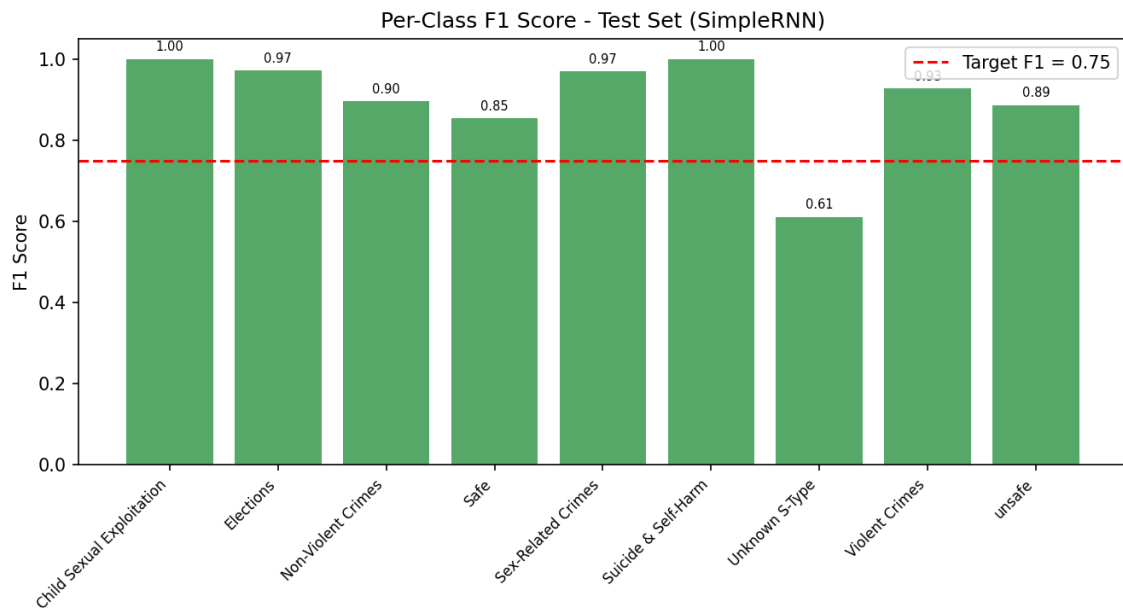
5. Training Curves



6. Confusion Matrix



7. Per-Class F1 Score



8. Conclusion

The Bidirectional SimpleRNN model achieved a weighted F1 score of 0.884 and a macro F1 score of 0.902 on the held-out test set, exceeding the required threshold of 0.75.